Group members:

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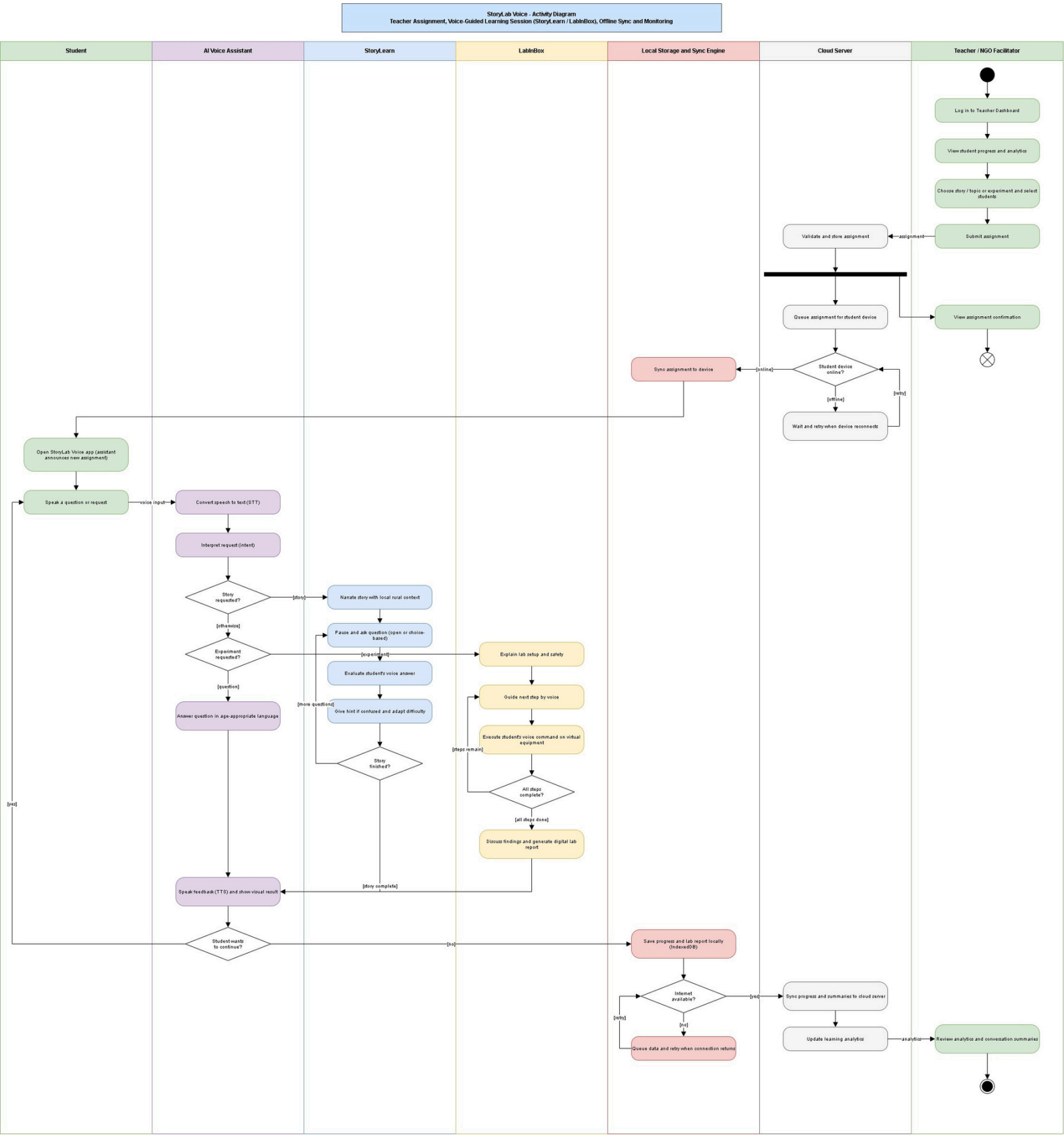
Ritesh Gavhane 202401100082

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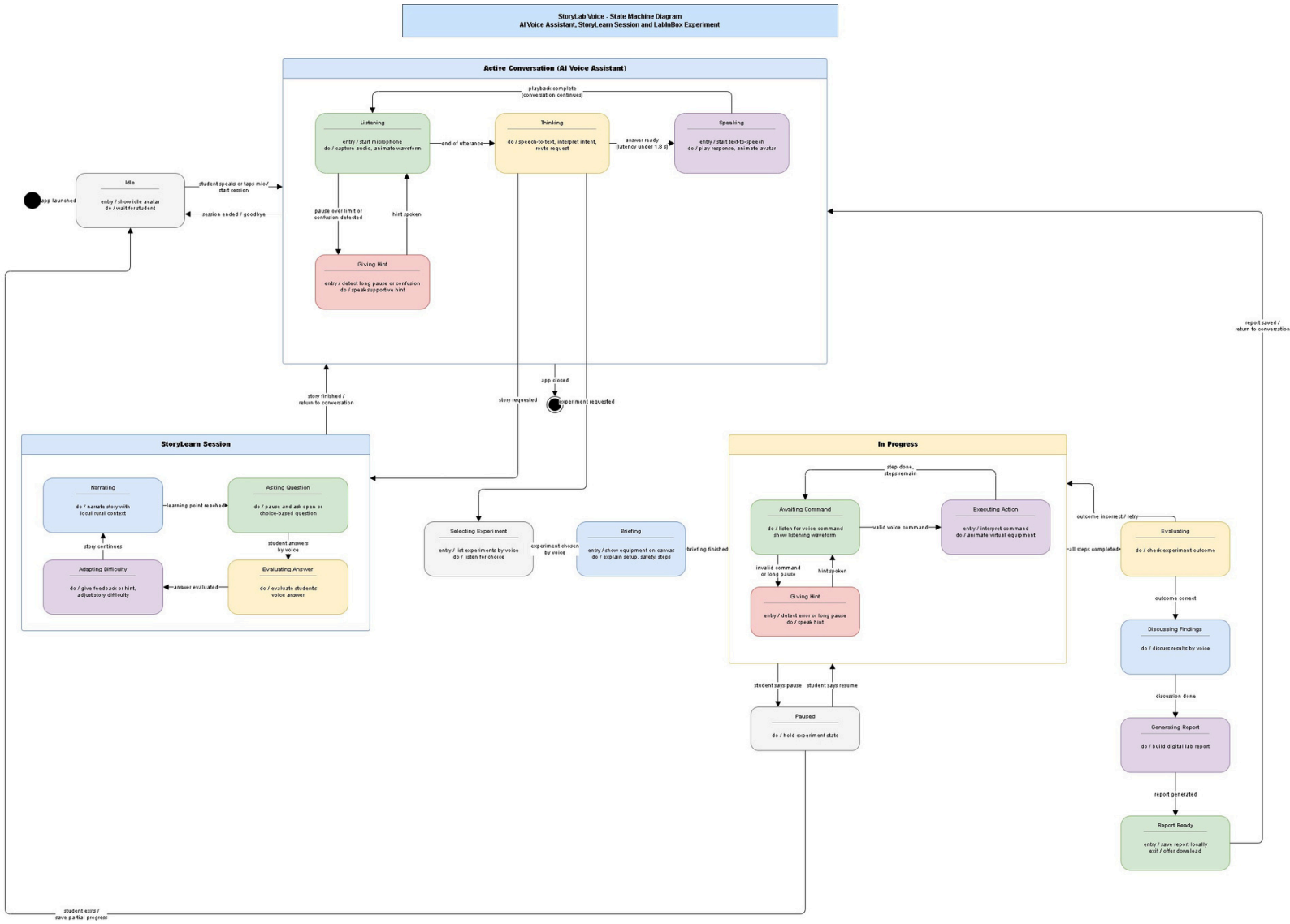
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PRACTICAL NO.05 Develop Workflow and State Models Using UML Activity and State Machine Diagrams for real world application.

Activity Diagram:



State Diagram:


StoryLab Voice - State Machine Diagram
AI Voice Assistant, StoryLearn Session and LabInBox Experiment
Active Conversation (AI Voice Assistant)
playback complete
[conversation continues]
Listening
entry / start microphone
do / capture audio, animate waveform
end of utterance
Thinking
do / speech-to-text, interpret intent, route request
answer ready
[latency under 1.8 s]
Speaking
entry / start text-to-speech
do / play response, animate avatar
app launched
Idle
entry / show idle avatar
do / wait for student
student speaks or taps mic / start session
session ended / goodbye
pause over limit or confusion detected
hint spoken
Giving Hint
entry / detect long pause or confusion
do / speak supportive hint
app closed
story requested
experiment requested
story finished / return to conversation
report saved / return to conversation
StoryLearn Session
Narrating
do / narrate story with local rural context
learning point reached
Asking Question
do / pause and ask open or choice-based question
story continues
student answers by voice
Adapting Difficulty
do / give feedback or hint, adjust story difficulty
answer evaluated
Evaluating Answer
do / evaluate studentfs voice answer
Selecting Experiment
entry / list experiments by voice
do / listen for choice
experiment chosen by voice
Briefing
entry / show equipment on canvas
do / explain setup, safety, steps
briefing finished
In Progress
step done, steps remain
Awaiting Command
do / listen for voice command
show listening waveform
valid voice command
Executing Action
entry / interpret command
do / animate virtual equipment
invalid command or long pause
hint spoken
Giving Hint
entry / detect error or long pause
do / speak hint
outcome incorrect / retry
all steps completed
Evaluating
do / check experiment outcome
outcome correct
Discussing Findings
do / discuss results by voice
discussion done
Generating Report
do / build digital lab report
report generated
Report Ready
entry / save report locally
exit / offer download
student says pause
student says resume
Paused
do / hold experiment state
student exits / save partial progress